PAS Data Processing Pipeline

Submission Information Package Creator for the Finnish Digital Preservation Service

Defining the Task

The Finnish Social Science Data Archive (FSD) is a national service resource that provides a single point of access to a wide range of digital research data for learning, teaching and research purposes. The Ministry of Education and Culture mandates Finnish Social Science Data Archive (FSD) to deposit its archived research data to the National Long Term Digital Preservation Service (NDP). This creates a need for tools to handle the painstaking tasks of collecting the required metadata, creating the SIP, and monitoring the package status.

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YHTEISKUNTATIETEELLINEN TIETOARKISTO
FINLANDS SAMHÄLLSVETENSKAPLIGA DATAARKIV
FINNISH SOCIAL SCIENCE DATA ARCHIVE

The FSD: http://www.fsd.uta.fi/en
Digital Preservation http://digitalpreservation.fi

File converter

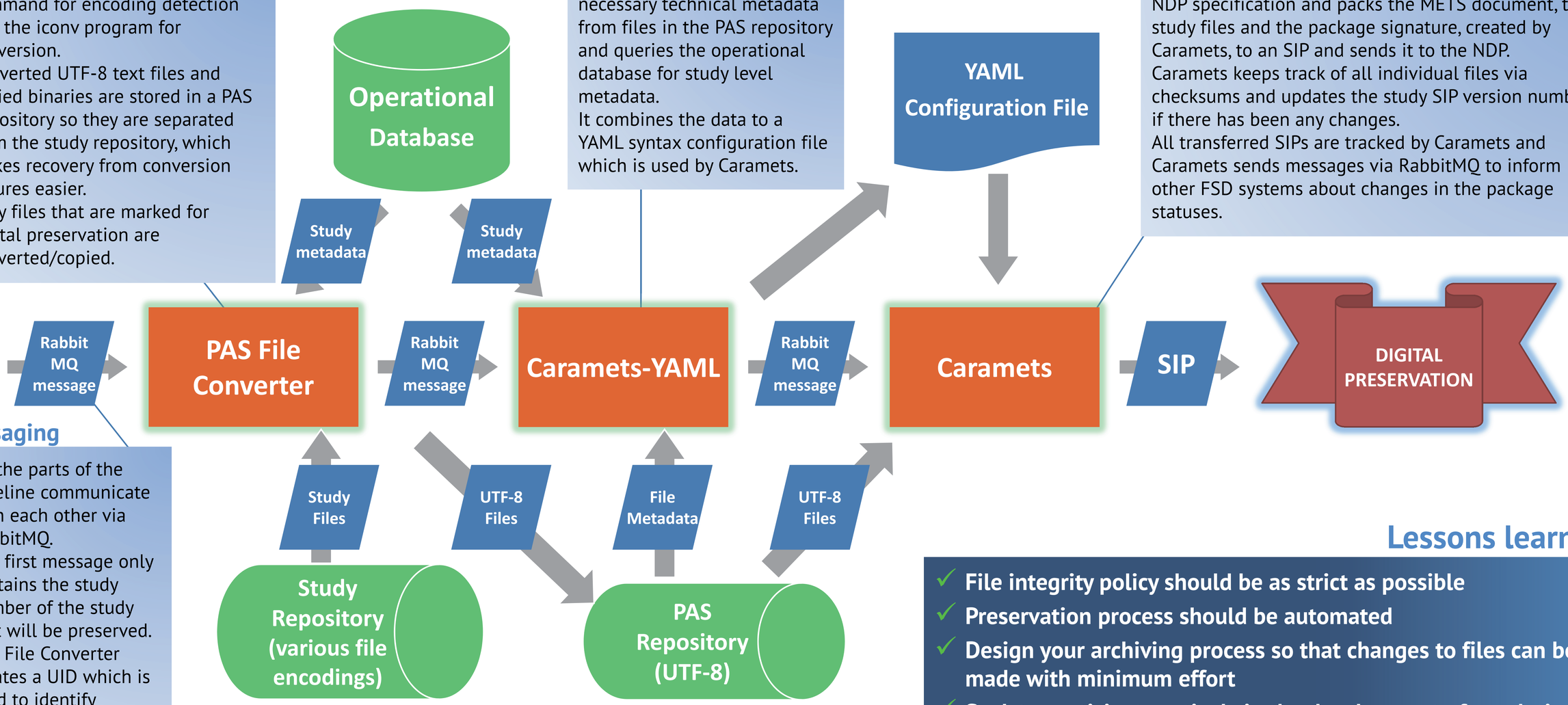
PAS file-converter uses the Unix file command for encoding detection and the iconv program for conversion.
Converted UTF-8 text files and copied binaries are stored in a PAS Repository so they are separated from the study repository, which makes recovery from conversion failures easier.
Only files that are marked for digital preservation are converted/copied.

Caramets-YAML

Caramets-YAML collects necessary technical metadata from files in the PAS repository and queries the operational database for study level metadata.
It combines the data to a YAML syntax configuration file which is used by Caramets.

Caramets

Caramets creates a METS document according to the NDP specification and packs the METS document, the study files and the package signature, created by Caramets, to an SIP and sends it to the NDP.
Caramets keeps track of all individual files via checksums and updates the study SIP version number if there has been any changes.
All transferred SIPs are tracked by Caramets and Caramets sends messages via RabbitMQ to inform other FSD systems about changes in the package statuses.



Messaging

All the parts of the pipeline communicate with each other via RabbitMQ.
The first message only contains the study number of the study that will be preserved.
PAS File Converter creates a UID which is used to identify specific SIPs in later messages.

Lessons learned

- ✓ File integrity policy should be as strict as possible
- ✓ Preservation process should be automated
- ✓ Design your archiving process so that changes to files can be made with minimum effort
- ✓ Seek to participate actively in the development of regulating higher (national) preservation specifications